

Telink

TL Matter SDK Release Note

Telink Semiconductor

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Information

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Revision History

- **1.1.0:** Initial release.

1 Overview

This document will introduce a matter customer solution based on our TLSR952x(2MB or 4MB), TL321x(2MB) and TL721x(2MB) platform.

Matter v1.4 was officially released in March 2025, and the certification process for version 1.4 has also commenced. Telink has been closely monitoring the updates to Matter, actively participating in all TE, SVE, and Plugfest events organized by the CSA. Following the official release of Matter v1.4, we have been conducting comprehensive internal testing and decided on the code version to be made available for customer use.

In this document, we will provide detailed recommendations for Telink customers regarding the preferred code versions for development. Additionally, we will furnish specific results from internal testing, including certification tests and ecosystem tests.

Note: All the tests based on Telink TLSR952x(2MB or 4MB), TL321x(2MB) and TL721x(2MB) platform.

2 Code Bases

2.1 Code Repository

Our TL_Matter SDK is Modified Matter SDK.

A complete Matter SDK consists of several repositories (Matter, Zephyr and so on). In this release, we recommend customers to use the following code bases for development on the Telink TLSR952x(2MB or 4MB) and TL321x(2MB) and TL721x(2MB) platform:

- **Matter repository:** <https://github.com/project-chip/connectedhomeip/tree/master>
Commit ID: af29048a0042a529e0ec37f575e09b41b08ce438
Patch: tl_matter_sdk_matter_v1_1_0.patch
- **Zephyr repository:** https://github.com/telink-semi/zephyr/tree/develop_v3.3
Commit ID: e20d103aa29b2ea9febdff81d16cf7336c3721d0
Patch: tl_matter_sdk_zephyr_v1_1_0.patch
- **hal_telink repository:** https://github.com/telink-semi/hal_telink/tree/develop
Commit ID: 541f19515a7796411ed68352ffd9718472212ca6
Patch: tl_matter_sdk_hal_v1_1_0.patch

- **modules_lib_openthread repository:** <https://github.com/zephyrproject-rtos/openthread/tree/main>

Commit ID: aabbce49c8c9721ed1ca22a7b326ae3a23aaae09

Patch: tl_matter_sdk_lib_openthread_v1_1_0.patch

2.2 Apply Patches



Please follow the steps below strictly in the given order.

- Apply the Matter patch

1. Copy the `tl_matter_sdk_matter_v1_1_0.patch` to the path of connectedhomeip folder.
2. Apply this patch

```
1 cd ~/connectedhomeip
2 git apply tl_matter_sdk_matter_v1_1_0.patch
```

- Apply the zephyr patch

1. Copy the `tl_matter_sdk_zephyr_v1_1_0.patch` to the path of zephyr folder.
2. Apply this patch and update hal, mcuboot

```
1 cd ~/zephyrproject/zephyr
2 git apply tl_matter_sdk_zephyr_v1_1_0.patch
3 cd ..
4 west update
```

- Apply the hal patch and fetch lib file

1. Copy the `tl_matter_sdk_hal_v1_1_0.patch` to the path of hal_telink folder.
2. Apply this patch

```
1 cd ~/zephyrproject/modules/hal/telink
2 git apply tl_matter_sdk_hal_v1_1_0.patch
3 cd ../../..
4 west blobs fetch hal_telink
```

- Apply modules_lib_openthread patch

1. Copy the `tl_matter_sdk_lib_openthread_v1_1_0.patch` to the path of `lib_openthread` folder.
2. Apply this patch

```
1 cd ~/zephyrproject/modules/lib/openthread
2 git apply tl_matter_sdk_lib_openthread_v1_1_0.patch
```

2.3 Change Device Type

By default, the lighting-app is a dimmable light, if you need to change the device type to extended color light, just need to change it via ZAP tool.

1. Run the ZAP tool

```
1 cd ~/connectedhomeip
2 source scripts/activate.sh
3 ./scripts/tools/zap/run_zaptool.sh examples/lighting-app/lighting-
  common/lighting-app.zap
```

2. Click Endpoint1 to see the device type

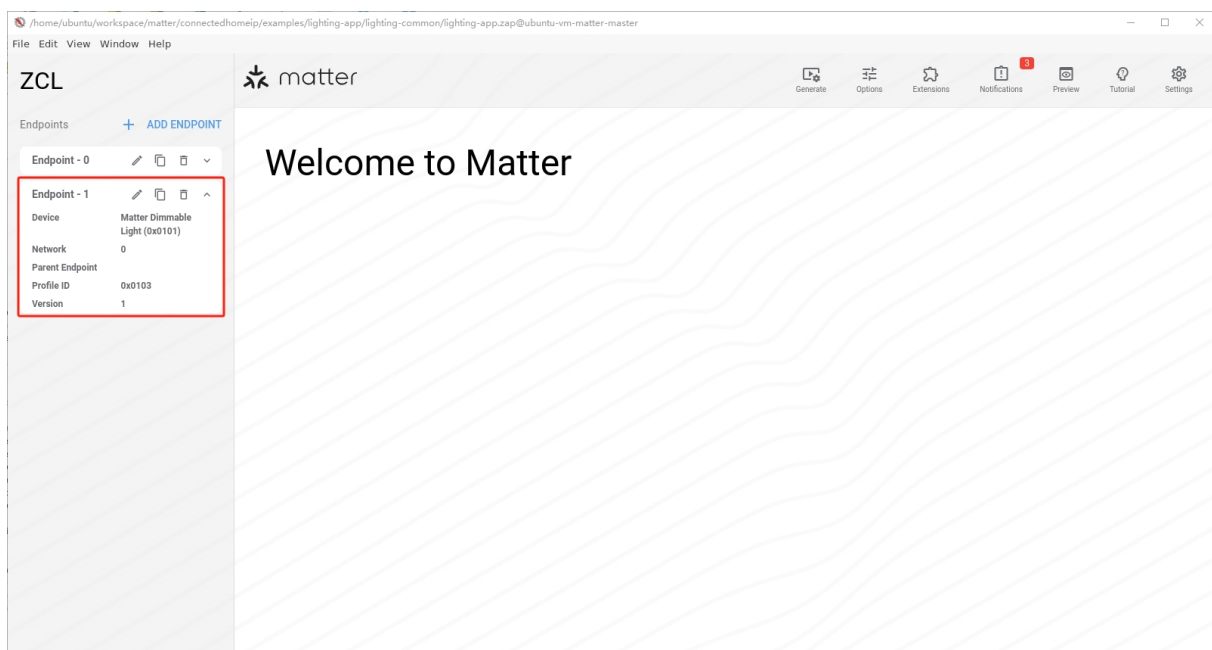


Figure 1: ZAP Endpoint1

3. Click **Edit** to change the device type in Endpoint1

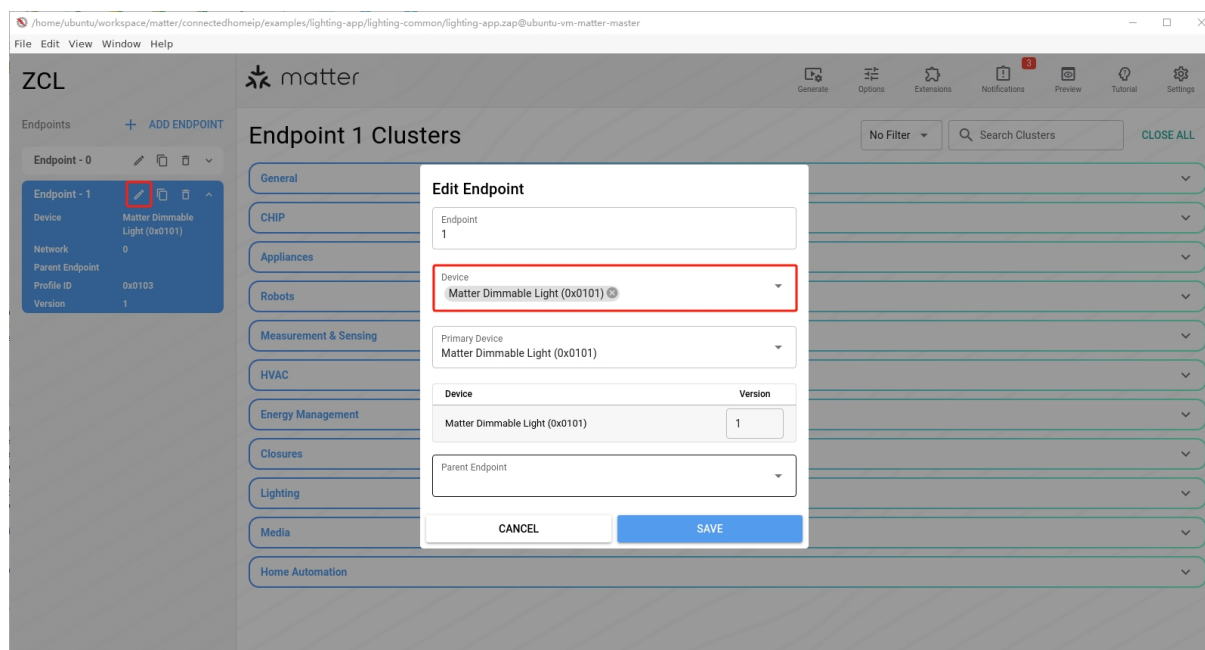


Figure 2: ZAP Change Device Type

4. Choose Matter Extended Color Light(0x10D)

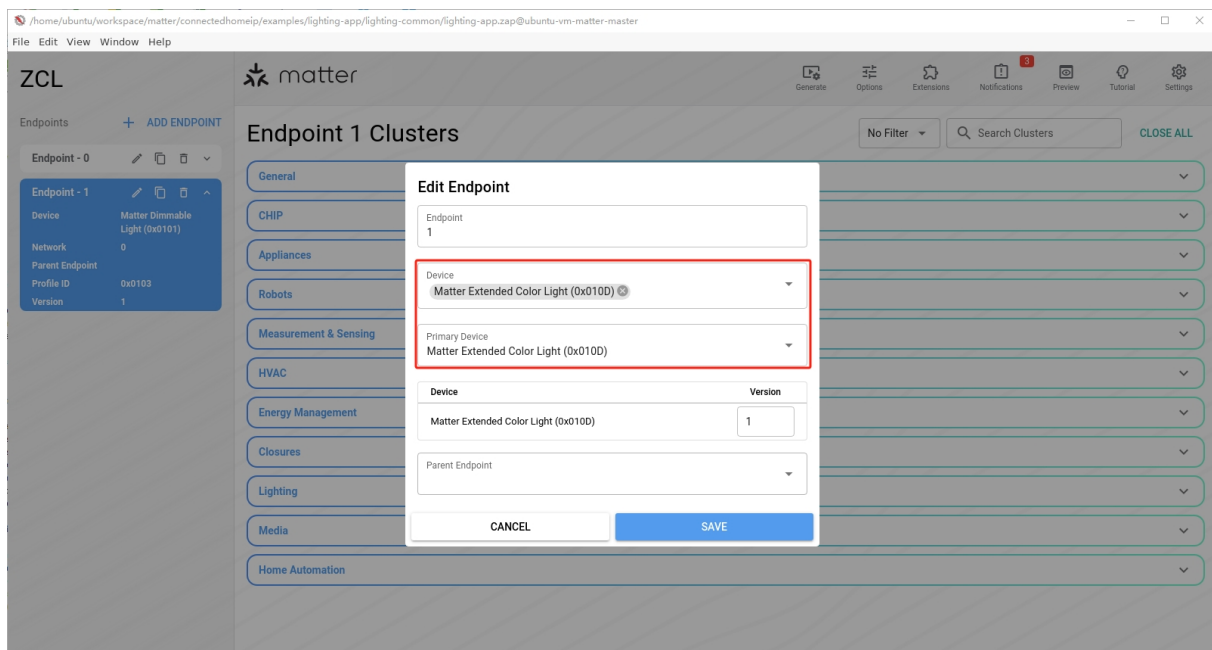


Figure 3: ZAP Choose Device Type

5. Save the changes

Click **File**–**save** in the top left corner of ZAP page or use **Ctrl+S** directly to save the changes.

6. Update `lighting-app.matter` file

```
1 cd ~/connectedhomeip
2 ./scripts/tools/zap/generate.py examples/lighting-app/lighting-
  common/lighting-app.zap
```

2.4 Generate the Firmware

1. Compile command:

```
1 cd ~/connectedhomeip
2 source scripts/activate.sh
3 cd examples/lighting-app/telink
4 rm -rf build/
5 west build -p auto -b tlr9528a
6 west build -p auto -b tl3218x
7 west build -p auto -b tl7218x
```

```
1 cd examples/light-switch-app/telink
2 rm -rf build/
3 west build -p auto -b tlr9528a_retention
4 west build -p auto -b tl3218x_retention
5 west build -p auto -b tl7218x_retention
```

If you want to get more information about how to set up a Matter environment, please check our [Telink Wiki page](#) and visit our developer's guide at the end of this webpage.

3 Matter Basic Functionality Test Results

We used the official TestHarness(version 2.11-beta1+fall2024) to conduct detailed tests on the basic functionality of the Matter device, including the following test items.

Test	Telink lighting-app	Telink light-switch-app
Factory Reset	PASS	PASS
Commissioning	PASS	PASS
OnOff Control	PASS	PASS
Reconnection	PASS	PASS
OTA Update	PASS	PASS

4 Matter Ecosystem Test Results

To simulate the everyday interactions users have with IoT devices, we conducted testing on specific device types within prominent commercial ecosystems such as Google, Apple, Samsung and Amazon.

Our simulated testing primarily includes the following tests:

1. **Commissioning Test** – Attempting commissioning operations using the same device types across the four ecosystems (Google, Apple, Samsung and Amazon).
2. **Interaction Test** – Controlling devices that have already been commissioned within each of the four ecosystems (Google, Apple, Samsung and Amazon).
3. **Stability Test** – Testing the duration for which devices that are already commissioned in each of the four ecosystems (Google, Apple, Samsung and Amazon) can maintain an online status.
4. **Reconnection Test** – Powering off devices that are already commissioned in each of the four ecosystems (Google, Apple, Samsung and Amazon), waiting for a certain period, and then powering them back on to test whether the devices can successfully reconnect.

4.1 Test Conditions

4.1.1 Telink Platform

We use Telink TLSR952x(2MB or 4MB), TL321x(2MB) and TL721x(2MB) EVB as the device under test, to test the behavior of different device types across four ecosystems.



Figure 4: TLSR952x-Development-Board-Front

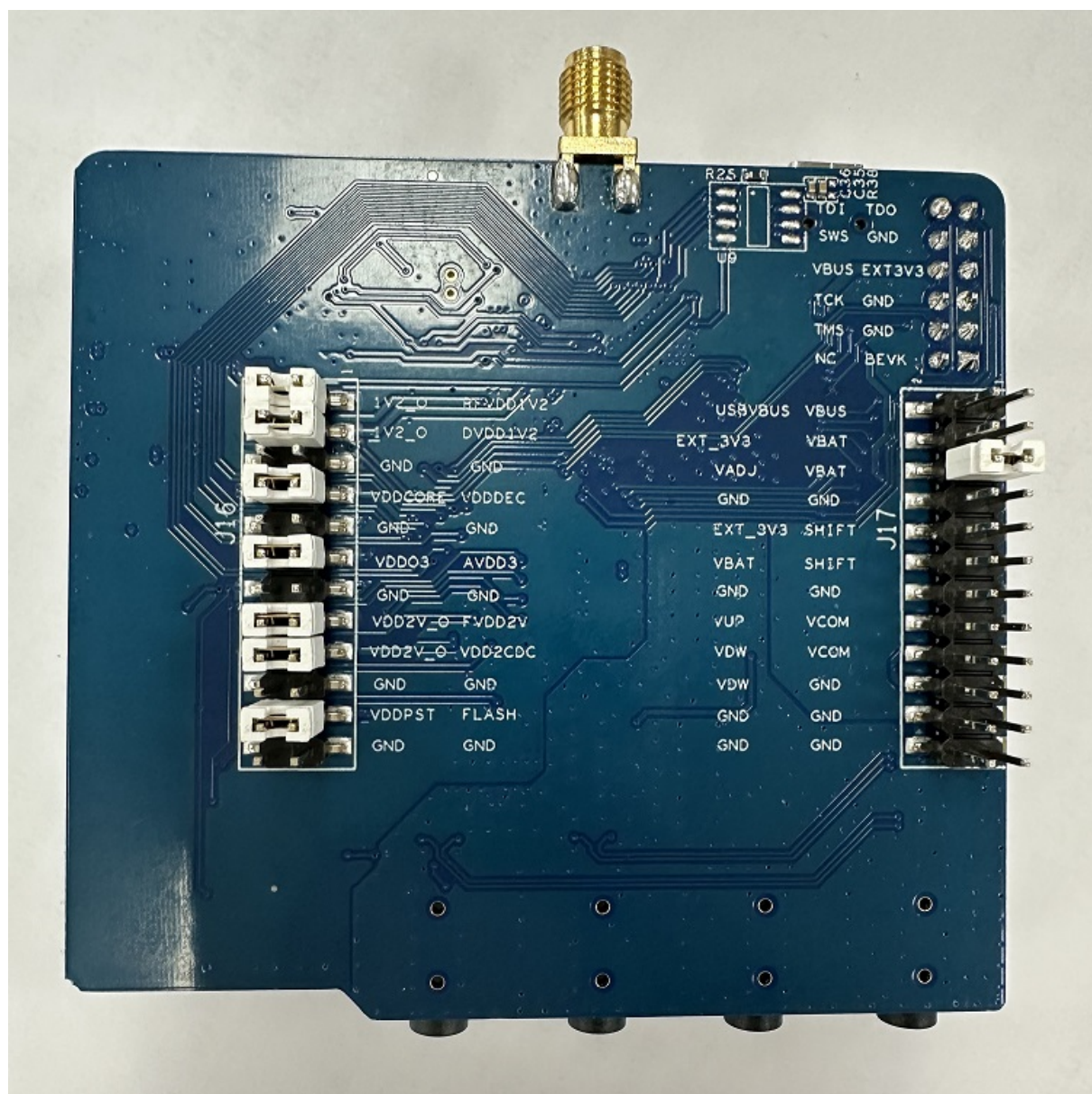
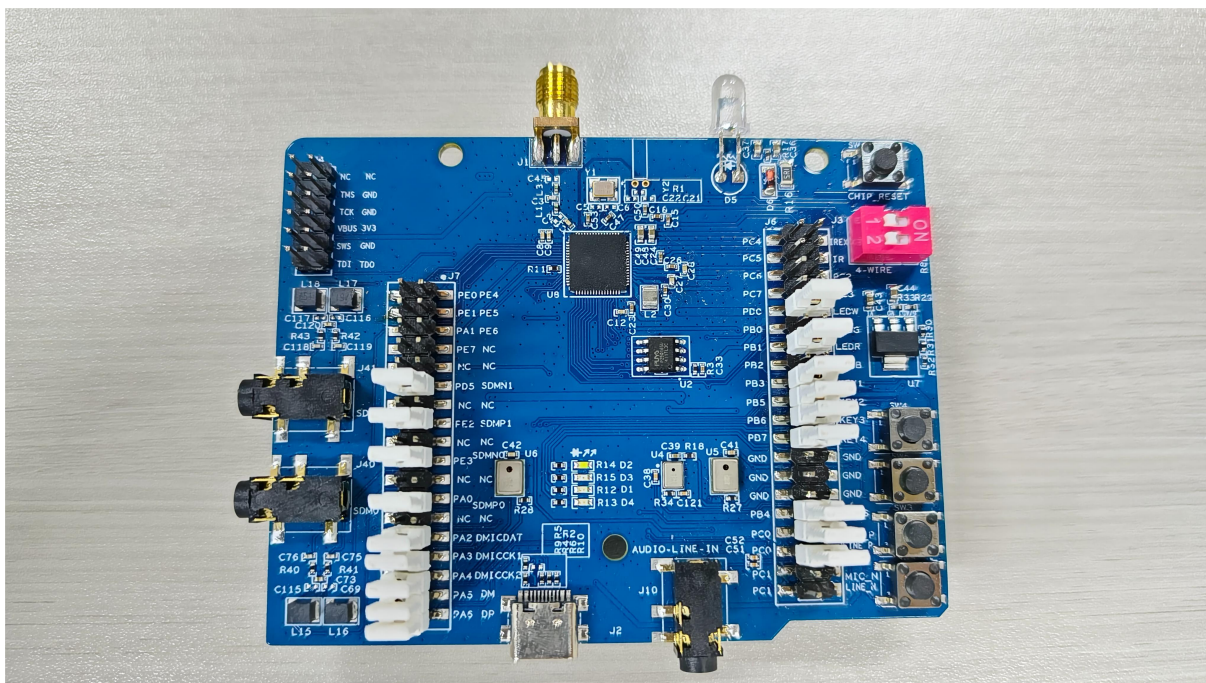
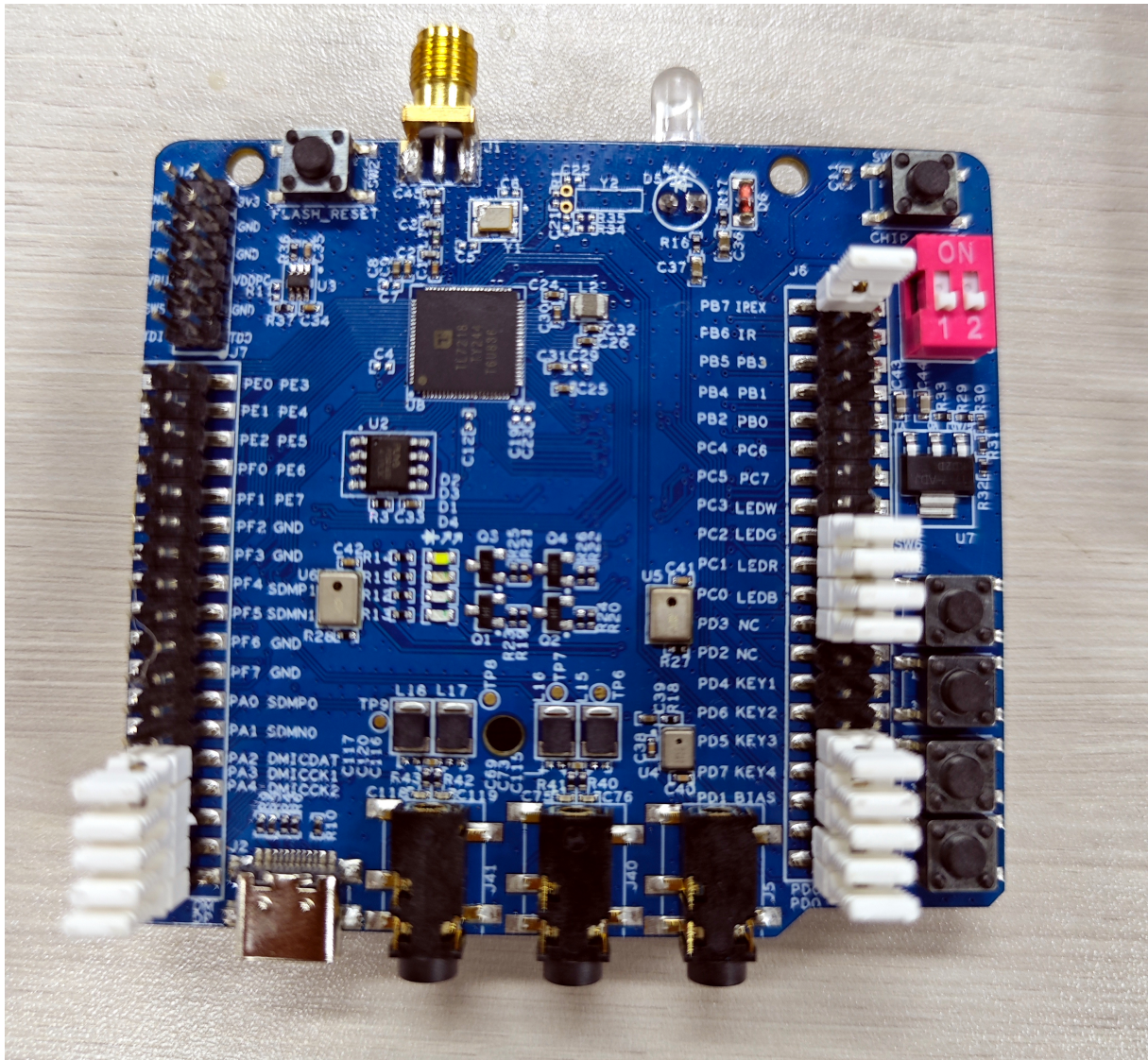


Figure 5: TLSR952x-Development-Board-Back

**Figure 6:** TL321x-Development-Board-Front**Figure 7:** TL321x-Development-Board-Back



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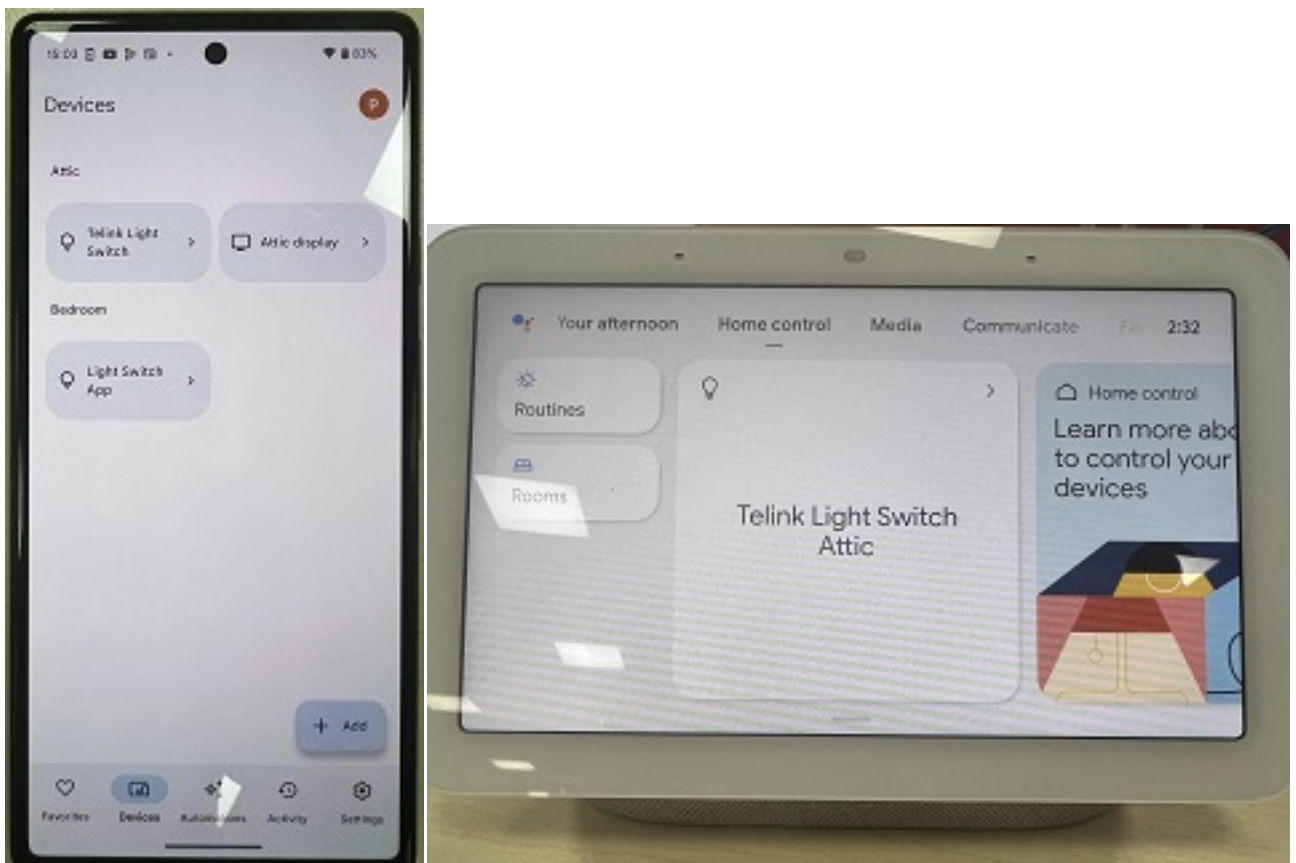
4.1.2 Apple Ecosystem

- iPhone 13
 - iOS version: iOS 17.5.1 (21F90)
- Apple Homepod mini
 - software version: 17.5 (21L569)



4.1.3 Google Ecosystem

- Pixel 6
 - Android version: 13
 - Google Play services: version 24.28.34 (190400-655627182)
- Google Nesthub (Gen 2)
 - Fuchsia Version: 18.20240225.3.63
 - Software Version: 58.102.2.613521407
 - Chromecast firmware version: 3.73.411796



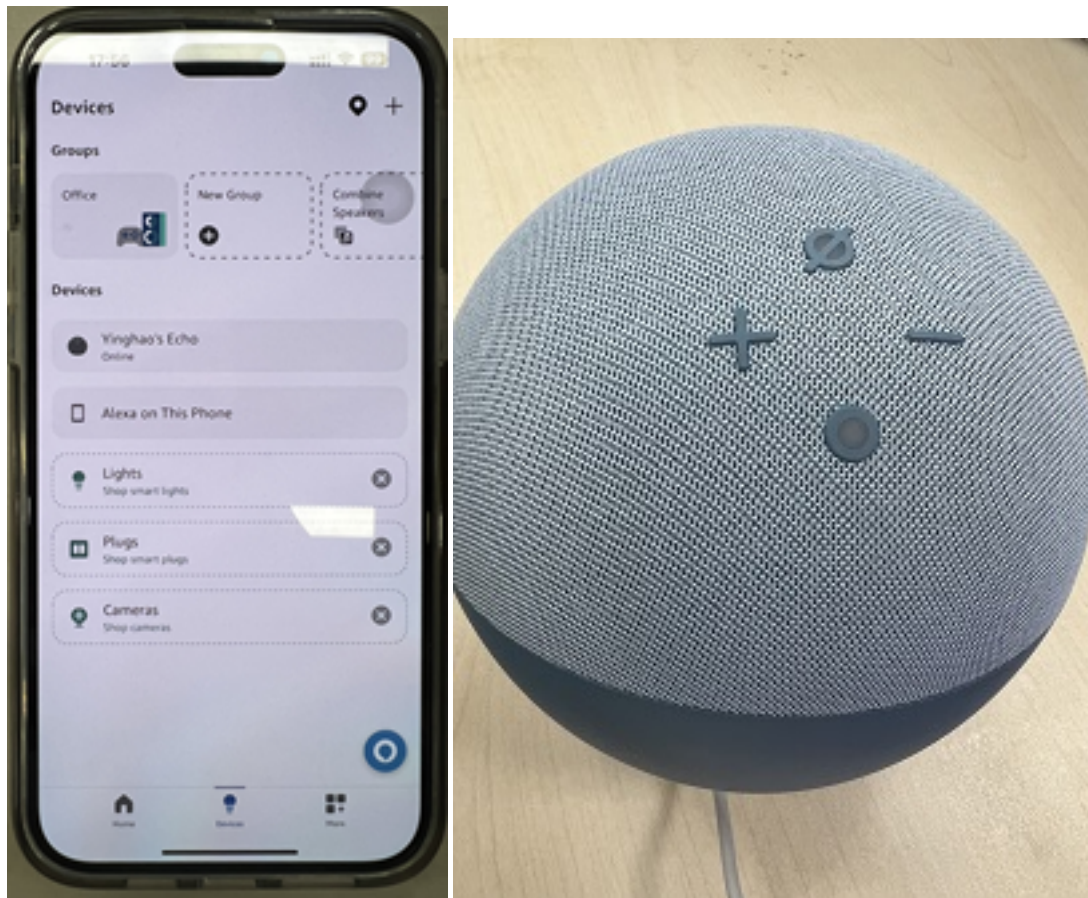
4.1.4 Samsung Ecosystem

- Pixel 6
 - Android version: 13
- Samsung SmartThings Station
 - Firmware version: P9501ZCU1AXF5



4.1.5 Amazon Ecosystem

- iPhone 14
 - iOS version: iOS 17.5.1 (21F90)
- Amazon Alexa Echo 4
 - Device Software Version: 0010369673092



4.2 Matter Ecosystem Test

4.2.1 Matter Commissioning Test

Test procedure:

1. Commission devices into different ecosystems.
2. Remove the commissioned devices.
3. Do factory reset.
4. Repeat Step 1 and 2 a total of 10 times continuously.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS	PASS
light-switch-app	PASS	PASS	PASS	PASS

4.2.2 Matter Interaction Test

Test procedure:

1. Commission devices into different ecosystems.
2. Control the devices that has been commissioned into ecosystems. For example, with regards to a light, we would attempt to control its on/off using a mobile app within the respective ecosystems.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS	PASS

4.2.3 Matter Stability Test

Test procedure:

1. Commission devices into different ecosystems.

2. Leaving the commissioned devices in place for 7 days (7 * 24 hours) and periodically checking whether the devices can be reliably controlled during this period.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	On-Going	On-Going	On-Going	On-Going
light-switch-app	On-Going	On-Going	On-Going	On-Going

4.2.4 Matter Reconnection Test

Test procedure:

1. Commission devices into different ecosystems.
2. Power off the devices, then power them on again. Afterward, check whether these devices can reconnect to the networks or not.
3. Repeat Step 1 and 2 a total of 10 times continuously.

Test results:

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS	PASS
light-switch-app	PASS	PASS	PASS	PASS

4.2.5 Multi Fabric Test

1. Apple Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Apple HomePod-mini.
- ii. Share the device with Google, Samsung and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Google NestHub	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS
light-switch-app	PASS	PASS	PASS

2. Google Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Google Nesthub.
- ii. Share the device with Apple, Samsung and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Apple HomePod-mini	Samsung SmartThings Station	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS
light-switch-app	PASS	PASS	PASS

3. Samsung Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Samsung SmartThings Station.
- ii. Share the device with Apple, Google and Amazon ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Google NestHub	Apple HomePod-mini	Amazon Alexa Echo
lighting-app	PASS	PASS	PASS
light-switch-app	PASS	PASS	PASS

4. Amazon Ecosystem Multi Fabric

Test procedure:

- i. Commission the device into Amazon Alexa.

- ii. Share the device with Apple, Google and Samsung ecosystems, make sure the device can be controlled by all four ecosystems.

Device Type	Google NestHub	Apple HomePod-mini	Samsung SmartThings Station
lighting-app	PASS	PASS	PASS
light-switch-app	PASS	PASS	PASS

5 Matter Certification Test Results

In order to bring Matter devices to market, it is imperative that they successfully undergo the Matter certification test, resulting in the issuance of a certificate by CSA. To achieve this, adherence to the latest Test Plans and test suites provided by the Matter community is mandatory.

To meet these requirements, we conducted a series of certification tests for the dual mode lighting app, selected a large number of automatic and python tests, and finally our device could pass all of these test cases. Here's the list of tested cases:



TestHarness version: 2.11-beta1+fall2024

Test	Telink lighting-app
TC-ACE-1.1	PASS
TC-ACE-1.2	PASS
TC-ACE-1.3	PASS
TC-ACE-1.4	PASS
TC-ACE-1.5	PASS
TC-ACE-1.6	PASS
TC-ACE-2.1	PASS
TC-ACE-2.2	PASS
TC-ACL-2.1	PASS
TC-ACL-2.2	PASS
TC-ACL-2.3	PASS
TC-ACL-2.6	PASS
TC-CGEN-2.1	PASS
TC-CGEN-2.4	PASS
TC-CNET-1.3	PASS
TC-CNET-4.6	PASS
TC-DESC-2.2	PASS

Test	Telink lighting-app
TC-DGGEN-2.3	PASS
TC-DGGEN-2.4	PASS
TC-DGGEN-3.1	PASS
TC-DGTHREAD-2.2	PASS
TC-DGTHREAD-2.3	PASS
TC-DT-1.1	PASS
TC-GRPKEY-2.1	PASS
TC-IDM-1.2	PASS
TC-IDM-1.4	PASS
TC-IDM-10.1	PASS
TC-IDM-10.3	PASS
TC-IDM-11.1	PASS
TC-LTIME-1.2	PASS
TC-OPCREDS-3.1	PASS
TC-SC-3.6	PASS
TC-SC-5.1	PASS
TC-SM-1.1	PASS
TC-SM-1.2	PASS
TC-ULABEL-2.1	PASS
TC-ULABEL-2.2	PASS
TC-ULABEL-2.3	PASS
TC-ULABEL-2.4	PASS
TC-G-2.1	PASS
TC-I-2.1	PASS
TC-OO-2.1	PASS
TC-OO-2.3	PASS
TC-OCC-2.1	PASS

Test	Telink lighting-app
TC-OCC-2.3	PASS
TC-RR-1.1	PASS
TC-CC-4.1	PASS
TC-CC-4.2	PASS
TC-CC-4.3	PASS
TC-CC-4.4	PASS
TC-CC-5.1	PASS
TC-CC-5.2	PASS
TC-CC-5.3	PASS
TC-CC-6.1	PASS
TC-CC-6.2	PASS
TC-CC-6.3	PASS
TC-CC-7.1	PASS
TC-CC-7.2	PASS
TC-CC-7.3	PASS
TC-CC-7.4	PASS
TC-CC-9.1	PASS
TC-CC-9.2	PASS
TC-CC-9.3	PASS